Field Vectoring with a Four-Channel Coil Array

Lecture notes on `lib/VectorField`

ESP32_PMW

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Abstract

We derive the algorithm in `VectorField::settingsFor`. Given a point $\mathbf{r}$, an orientation R and a field strength b, it returns each channel's carrier duty and phase so that the field at $\mathbf{r}$ rotates in a circle of radius b about the axis $R\hat{\mathbf{z}}$. We then derive the cost of tilting the axis, the effect of a twist about the axis, the effect of calibration errors, and the maximum field the current limits allow for each axis. The longer write-up is `controller/control/theory.md` §12 and §27.

1 Problem statement

The rotor's spin axis follows the rotation axis of the field at its position, and the rotor's thrust points along its spin axis. To tilt the thrust we must set the field's rotation axis.

Per channel, the firmware sets a carrier duty, which sets the current amplitude, and a phase. All four channels share one frequency. We want the duties and phases that make, at a point $\mathbf{r}$, a circular field of strength b rotating about a unit axis $\hat{\mathbf{n}}$.

Notation. Bold symbols are vectors, hats are unit vectors, and j is the imaginary unit ($\mathrm{j}^2 = -1$), which frees i for currents. Id_3 is the 3×3 identity. $\mathbf{r} = (x, y, z)$ is a point in the array frame.

2 Model of the array

Geometry. The four channels A, B, C, D sit at azimuths 0°, 90°, 180° and 270° (measured 2026-08-29). Each channel is a 2×2 block of air-core coils, centred 37 mm from the array axis, with 21 mm between coil centroids. So channel A's coils sit at $(37 \pm 10.5,\ \pm 10.5,\ 0)$ mm. Every coil has radius $a = 10.5$ mm, $N = 650$ turns and its normal along $+\hat{\mathbf{z}}$. The coil plane is $z = 0$. The robot flies at z between about 15 and 25 mm, near the array axis.

Currents. All four coil currents are sinusoids at one shared drive frequency f, so they all repeat with period $1/f$. We call one such period a *drive cycle*. To say where we are inside a drive cycle, we define the *drive angle*

$$\theta(t) = 2\pi f t, \tag{1}$$

which advances by 2π every drive cycle.

Channel k's current has its own amplitude and its own phase offset from θ. We describe it by a complex number I_k, the channel's *phasor*, and write

$$i_k(t) = \mathrm{Re}\big(I_k e^{\mathrm{j}\theta}\big) = \mathrm{Re}\big(|I_k|\, e^{\mathrm{j}(\theta + \arg(I_k))}\big) = |I_k| \cos\big(\theta + \arg(I_k)\big). \tag{2}$$

Here $|I_k| = \sqrt{\mathrm{Re}(I_k)^2 + \mathrm{Im}(I_k)^2}$ is the magnitude of I_k, its length in the complex plane, and it is the current amplitude in amps. $\arg(I_k)$ is the angle of I_k in the complex plane, measured from the positive real axis. The current peaks at $\theta = -\arg(I_k)$, so $\arg(I_k)$ is how far channel k runs ahead of θ. The four phasors form $\mathbf{I} = (I_A, I_B, I_C, I_D) \in \mathbb{C}^4$.

Field per amp. We replace each coil by a point dipole. Let $\mathbf{c}$ be the coil's centroid; we place the dipole at $\mathbf{c}$ and give it the coil's moment per amp, $\mathbf{m} = N\pi a^2\,\hat{\mathbf{z}}$. The robot sits about 35 to 55 mm from the coil centroids, several coil radii away, where the dipole term dominates the field of a loop. Section 7.2 compares this model with the exact loop model. A channel's field per amp at $\mathbf{r}$ is the sum over its four coils,

$$\mathbf{b}_k(\mathbf{r}) = \frac{\mu_0}{4\pi} \sum_{\mathbf{c}\in k} \left[\frac{3(\mathbf{m}\cdot\mathbf{d})\,\mathbf{d}}{|\mathbf{d}|^5} - \frac{\mathbf{m}}{|\mathbf{d}|^3} \right], \qquad \mathbf{d} = \mathbf{r} - \mathbf{c}. \tag{3}$$

We stack the four columns into the *channel matrix*

$$B(\mathbf{r}) = [\,\mathbf{b}_A \;\; \mathbf{b}_B \;\; \mathbf{b}_C \;\; \mathbf{b}_D\,] \in \mathbb{R}^{3\times 4} \quad \text{(T/A)}. \tag{4}$$

Column k is the field that one amp in channel k makes at $\mathbf{r}$. Fields superpose, so currents $\mathbf{i}$ make the field $B\mathbf{i}$. At $f \le 250$ Hz the wavelength is kilometres and nothing nearby conducts, so the field is quasi-static and B does not depend on f. From here on, B is evaluated at the robot's position.

3 Field over one drive cycle

Substituting the currents into $\mathbf{B} = B\mathbf{i}$,

$$\mathbf{B}(\theta) = \sum_k \mathbf{b}_k\, i_k = \mathrm{Re}\big(B\mathbf{I}\, e^{\mathrm{j}\theta}\big) = B\ \mathrm{Re}(\mathbf{I})\,\cos\theta - B\ \mathrm{Im}(\mathbf{I})\,\sin\theta. \tag{5}$$

In the last step we expand $e^{\mathrm{j}\theta} = \cos\theta + \mathrm{j}\sin\theta$ and use that B is real. We name the two coefficients:

$$\boxed{\mathbf{B}(\theta) = \mathbf{u}\cos\theta + \mathbf{v}\sin\theta, \qquad \mathbf{u} = B\ \mathrm{Re}(\mathbf{I}), \qquad \mathbf{v} = -B\ \mathrm{Im}(\mathbf{I})} \tag{6}$$

$\mathbf{u}$ is the field at $\theta = 0$ and $\mathbf{v}$ is the field at $\theta = \pi/2$. The tip of $\mathbf{B}$ traces an ellipse in the plane spanned by $\mathbf{u}$ and $\mathbf{v}$.

Rotation axis. We differentiate, $d\mathbf{B}/d\theta = -\mathbf{u}\sin\theta + \mathbf{v}\cos\theta$, and take the cross product:

$$\mathbf{B} \times \frac{d\mathbf{B}}{d\theta} = \cos^2\theta\,(\mathbf{u}\times\mathbf{v}) - \sin^2\theta\,(\mathbf{v}\times\mathbf{u}) = \mathbf{u}\times\mathbf{v}. \tag{7}$$

The result does not depend on θ, so the field turns about the fixed direction

$$\hat{\mathbf{n}} = \frac{\mathbf{u}\times\mathbf{v}}{\|\mathbf{u}\times\mathbf{v}\|}, \tag{8}$$

by the right-hand rule. The sign of $\hat{\mathbf{n}}$ gives the spin sense.

Circular field. $|\mathbf{B}(\theta)|^2 = |\mathbf{u}|^2\cos^2\theta + |\mathbf{v}|^2\sin^2\theta + 2\,\mathbf{u}\cdot\mathbf{v}\,\sin\theta\cos\theta$. This equals b^2 for every θ if and only if $|\mathbf{u}| = |\mathbf{v}| = b$ and $\mathbf{u}\cdot\mathbf{v} = 0$. Then the field has constant magnitude b and turns at the constant rate $2\pi f$.

4 Commanded orientation

A circular field of radius b is fixed by $\mathbf{u}$ and $\mathbf{v}$: two perpendicular vectors of length b. Their directions $\hat{\mathbf{e}}_1 = \mathbf{u}/b$, $\hat{\mathbf{e}}_2 = \mathbf{v}/b$ and $\hat{\mathbf{n}} = \hat{\mathbf{e}}_1 \times \hat{\mathbf{e}}_2$ form a right-handed orthonormal frame, which is the columns of a rotation matrix. So the caller supplies a rotation $R \in SO(3)$, or a unit quaternion that the library converts to one, and we read its columns as

$$\begin{aligned} \hat{\mathbf{e}}_1 &= R\hat{\mathbf{x}} && \text{field direction at } \theta = 0, \\ \hat{\mathbf{e}}_2 &= R\hat{\mathbf{y}} && \text{field direction at } \theta = \pi/2, \\ \hat{\mathbf{n}} &= R\hat{\mathbf{z}} && \text{rotation axis.} \end{aligned} \tag{9}$$

The target is $\mathbf{u} = b\,\hat{\mathbf{e}}_1$, $\mathbf{v} = b\,\hat{\mathbf{e}}_2$. Then $\mathbf{u} \times \mathbf{v} = b^2\,\hat{\mathbf{e}}_1 \times \hat{\mathbf{e}}_2 = b^2\hat{\mathbf{n}}$, because $\det(R) = +1$. $R = \mathrm{Id}_3$ asks for a field that starts along $+x$ and rotates about $+z$.

Twist about the axis. R has three angles and the axis $\hat{\mathbf{n}}$ has two. We find what the third does. We twist R by an angle ψ about its own axis, $R' = R\,R_z(\psi)$. The new columns are

$$\hat{\mathbf{e}}_1' = \cos\psi\,\hat{\mathbf{e}}_1 + \sin\psi\,\hat{\mathbf{e}}_2, \qquad \hat{\mathbf{e}}_2' = -\sin\psi\,\hat{\mathbf{e}}_1 + \cos\psi\,\hat{\mathbf{e}}_2, \qquad \hat{\mathbf{n}}' = \hat{\mathbf{n}}. \tag{10}$$

The solution in Section 5 depends on R only through $\hat{\mathbf{e}}_1 - \mathrm{j}\hat{\mathbf{e}}_2$, which transforms as

$$\hat{\mathbf{e}}_1' - \mathrm{j}\hat{\mathbf{e}}_2' = (\cos\psi + \mathrm{j}\sin\psi)\,\hat{\mathbf{e}}_1 + (\sin\psi - \mathrm{j}\cos\psi)\,\hat{\mathbf{e}}_2 = e^{\mathrm{j}\psi}\,(\hat{\mathbf{e}}_1 - \mathrm{j}\hat{\mathbf{e}}_2), \tag{11}$$

since $\sin\psi - \mathrm{j}\cos\psi = -\mathrm{j}e^{\mathrm{j}\psi}$. The solution is linear in this vector, so $\mathbf{I}' = e^{\mathrm{j}\psi}\mathbf{I}$. Each amplitude $|I_k|$ is unchanged and each phase advances by ψ. In the field,

$$\mathbf{B}'(\theta) = \mathrm{Re}\big(B\,e^{\mathrm{j}\psi}\mathbf{I}\,e^{\mathrm{j}\theta}\big) = \mathbf{B}(\theta + \psi), \tag{12}$$

the same field shifted in time by $\psi/(2\pi f)$. The twist sets the time origin and nothing else.

Continuity of the twist. The firmware applies a new phase at once, so a step $\Delta\psi$ in the twist steps the field direction by $\Delta\psi$. A synchronous rotor that lags the field by φ makes torque $\propto \sin\varphi$, which peaks at $\varphi = \pi/2$, so it slips once $\varphi + \Delta\psi > \pi/2$. The twist must therefore change continuously along a trajectory. If the API took only $\hat{\mathbf{n}}$, the library would have to choose $\hat{\mathbf{e}}_1$ as a function of $\hat{\mathbf{n}}$, a tangent vector field on S^2, and by the hairy-ball theorem every such choice is discontinuous somewhere. So the caller supplies the twist, and a quaternion slerp between commands keeps it continuous.

5 Inversion

5.1 Real linear system

The field is nonlinear in amplitude and phase, because of the cosine, but linear in the real and imaginary parts of $\mathbf{I}$. We set

$$\mathbf{x} = \mathrm{Re}(\mathbf{I}), \qquad \mathbf{y} = -\,\mathrm{Im}(\mathbf{I}) \qquad (\mathbf{x}, \mathbf{y} \in \mathbb{R}^4), \tag{13}$$

so that

$$i_k(t) = \mathrm{Re}\,\big((x_k - \mathrm{j}y_k)(\cos\theta + \mathrm{j}\sin\theta)\big) = x_k\cos\theta + y_k\sin\theta. \tag{14}$$

$\mathbf{x}$ is the cosine part of each channel's current and $\mathbf{y}$ the sine part. Section 3 gives $\mathbf{u} = B\mathbf{x}$ and $\mathbf{v} = B\mathbf{y}$, and the target is

$$\begin{bmatrix} B & 0 \\ 0 & B \end{bmatrix} \begin{bmatrix} \mathbf{x} \\ \mathbf{y} \end{bmatrix} = \begin{bmatrix} b\,\hat{\mathbf{e}}_1 \\ b\,\hat{\mathbf{e}}_2 \end{bmatrix}. \tag{15}$$

That is six equations in eight unknowns. The matrix is block diagonal, so it splits into two independent problems, $B\mathbf{x} = b\hat{\mathbf{e}}_1$ and $B\mathbf{y} = b\hat{\mathbf{e}}_2$, each three equations in four unknowns. Wherever B has rank 3, each has a one-parameter family of solutions, and we need a rule to pick one.

5.2 Minimum-heat solution

We pick the solution that dissipates the least power, because coil temperature is the binding limit on the rig. The mean ohmic power in channel k is $R_s\langle i_k^2\rangle = \frac{R_s}{2}(x_k^2 + y_k^2)$, with R_s the series resistance. Summed over channels it is $\frac{R_s}{2}\big(\|\mathbf{x}\|^2 + \|\mathbf{y}\|^2\big)$. The two blocks share no unknowns, so we minimise each norm separately. For one block with target $\mathbf{t}$ we solve

$$\min_{\mathbf{x}} \ \tfrac{1}{2}\|\mathbf{x}\|^2 \quad \text{subject to} \quad B\mathbf{x} = \mathbf{t}. \tag{16}$$

We form the Lagrangian $\mathcal{L} = \frac{1}{2}\|\mathbf{x}\|^2 - \boldsymbol{\lambda}^\top(B\mathbf{x} - \mathbf{t})$. Setting $\partial\mathcal{L}/\partial\mathbf{x} = 0$ gives $\mathbf{x} = B^\top\boldsymbol{\lambda}$. Substituting this into the constraint, we get $BB^\top\boldsymbol{\lambda} = \mathbf{t}$, so

$$\mathbf{x} = P\,\mathbf{t}, \qquad P = B^\top(BB^\top)^{-1} \in \mathbb{R}^{4\times 3}, \tag{17}$$

the pseudo-inverse of B. This is the global minimum. Any other solution is $\mathbf{x} + \boldsymbol{\eta}$ with $B\boldsymbol{\eta} = 0$, and $\mathbf{x}^\top\boldsymbol{\eta} = \boldsymbol{\lambda}^\top B\boldsymbol{\eta} = 0$, so $\|\mathbf{x} + \boldsymbol{\eta}\|^2 = \|\mathbf{x}\|^2 + \|\boldsymbol{\eta}\|^2 \geq \|\mathbf{x}\|^2$.

We apply P to both blocks and recombine $\mathbf{I} = \mathbf{x} - \mathrm{j}\mathbf{y}$:

$$\boxed{\mathbf{I} = P(\mathbf{u} - \mathrm{j}\mathbf{v}) = b\,P\,(\hat{\mathbf{e}}_1 - \mathrm{j}\hat{\mathbf{e}}_2)} \tag{18}$$

It is exact wherever B has rank 3, because then $BP = BB^\top(BB^\top)^{-1} = \mathrm{Id}_3$. The firmware inverts the 3×3 Gram matrix $G = BB^\top$ as $\mathrm{adj}(G)/\det(G)$.

5.3 Heat as a function of the axis

$P^\top P = G^{-1}BB^\top G^{-1} = G^{-1}$, so $\|P\mathbf{t}\|^2 = \mathbf{t}^\top G^{-1}\mathbf{t}$ and

$$\|\mathbf{x}\|^2 + \|\mathbf{y}\|^2 = b^2\big(\hat{\mathbf{e}}_1^\top G^{-1}\hat{\mathbf{e}}_1 + \hat{\mathbf{e}}_2^\top G^{-1}\hat{\mathbf{e}}_2\big) = b^2\big(\operatorname{tr} G^{-1} - \hat{\mathbf{n}}^\top G^{-1}\hat{\mathbf{n}}\big). \tag{19}$$

The second step writes $\mathbf{t}^\top G^{-1}\mathbf{t} = \mathrm{tr}(G^{-1}\mathbf{t}\mathbf{t}^\top)$ and uses $\hat{\mathbf{e}}_1\hat{\mathbf{e}}_1^\top + \hat{\mathbf{e}}_2\hat{\mathbf{e}}_2^\top = \mathrm{Id}_3 - \hat{\mathbf{n}}\hat{\mathbf{n}}^\top$. The heat depends on $\hat{\mathbf{n}}$ and not on the twist. It is minimised by the $\hat{\mathbf{n}}$ that maximises $\hat{\mathbf{n}}^\top G^{-1}\hat{\mathbf{n}}$, the eigenvector of G with the smallest eigenvalue σ_3^2.

5.4 Solution on the array axis

We write the singular value decomposition $B = \sum_j \sigma_j\,\mathbf{a}_j\mathbf{w}_j^\top$, with field directions $\mathbf{a}_j \in \mathbb{R}^3$ and drive patterns $\mathbf{w}_j \in \mathbb{R}^4$. Then $P = \sum_j \sigma_j^{-1}\mathbf{w}_j\mathbf{a}_j^\top$: field along $\mathbf{a}_j$ is made by the pattern $\mathbf{w}_j$ at $1/\sigma_j$ amps per tesla.

On the array axis, $\mathbf{r} = (0, 0, z)$, the four-fold symmetry fixes the form of B. Channel A's coils lie at $+x$. A $+\hat{\mathbf{z}}$ dipole at $(+x_c, 0, 0)$ gives $\mathbf{d} = (-x_c, 0, z)$ and an x-component $\propto 3z(-x_c)/|\mathbf{d}|^5 < 0$, so channel A's field at the axis points toward $-x$. Rotating by 90° for each channel,

$$B = \begin{bmatrix} -\beta & 0 & \beta & 0 \\ 0 & -\beta & 0 & \beta \\ \gamma & \gamma & \gamma & \gamma \end{bmatrix}, \qquad G = BB^\top = \mathrm{diag}(2\beta^2,\ 2\beta^2,\ 4\gamma^2), \tag{20}$$

where $\beta > 0$ is one channel's inward field per amp and γ its vertical field per amp. The off-diagonal entries of G vanish because $\gamma(-\beta + \beta) = 0$. G is diagonal, so the $\mathbf{a}_j$ are the coordinate axes, $\sigma_j = \sqrt{G_{jj}}$, and $\mathbf{w}_j = B^\top \mathbf{a}_j/\sigma_j$:

$\mathbf{a}_j$	σ_j	$\mathbf{w}_j$
$\hat{\mathbf{x}}$	$\sqrt{2}\,\beta$	$(-1, 0, 1, 0)/\sqrt{2}$, A against C
$\hat{\mathbf{y}}$	$\sqrt{2}\,\beta$	$(0, -1, 0, 1)/\sqrt{2}$, B against D
$\hat{\mathbf{z}}$	2γ	$(1, 1, 1, 1)/2$, the common mode
none	0	$(1, -1, 1, -1)/2$, the null mode

The null mode satisfies $B\mathbf{w} = 0$: it makes no field at $\mathbf{r}$ and the minimum-norm solution contains none of it. At $z = 20.1\,\text{mm}$, $\sigma = (2.42,\ 2.42,\ 0.36)\,\text{mT/A}$, so vertical field costs $2.42/0.36 = 6.7$ times more current per tesla than horizontal field.

The rows of $P = B^\top G^{-1}$ are

$$\mathbf{p}_A = \Big(-\tfrac{1}{2\beta},\, 0,\, \tfrac{1}{4\gamma}\Big), \quad \mathbf{p}_B = \Big(0,\, -\tfrac{1}{2\beta},\, \tfrac{1}{4\gamma}\Big), \quad \mathbf{p}_C = \Big(\tfrac{1}{2\beta},\, 0,\, \tfrac{1}{4\gamma}\Big), \quad \mathbf{p}_D = \Big(0,\, \tfrac{1}{2\beta},\, \tfrac{1}{4\gamma}\Big), \tag{21}$$

and $I_k = b\,\mathbf{p}_k \cdot (\hat{\mathbf{e}}_1 - \mathrm{j}\hat{\mathbf{e}}_2)$.

Vertical axis. $R = \mathrm{Id}_3$ gives $\hat{\mathbf{e}}_1 - \mathrm{j}\hat{\mathbf{e}}_2 = (1, -\mathrm{j}, 0)$, so

$$\mathbf{I} = \frac{b}{2\beta}\,(-1,\ \mathrm{j},\ 1,\ -\mathrm{j}). \tag{22}$$

The four phasors have equal magnitude and angles $180°$, $90°$, $0°$ and $-90°$: each channel lags the previous one by $90°$.

Tilted axis. We tilt by α toward $+x$, taking R as a rotation about $\hat{\mathbf{y}}$: $\hat{\mathbf{e}}_1 = (\cos\alpha, 0, -\sin\alpha)$, $\hat{\mathbf{e}}_2 = \hat{\mathbf{y}}$, $\hat{\mathbf{n}} = (\sin\alpha, 0, \cos\alpha)$. Then

$$I_k = b\big(p_{k,x}\cos\alpha - \mathrm{j}\,p_{k,y} - p_{k,z}\sin\alpha\big), \qquad \mathbf{I} = \frac{b}{2\beta}\,(-\cos\alpha,\ \mathrm{j},\ \cos\alpha,\ -\mathrm{j}) \ - \ \frac{b\sin\alpha}{4\gamma}\,(1,1,1,1). \tag{23}$$

The tilt puts a vertical component $-b\sin\alpha$ into $\mathbf{u}$, and only the common mode makes vertical field, so every channel gains the same real term. Its size relative to the A–C drive is

$$\frac{\sin\alpha/(4\gamma)}{\cos\alpha/(2\beta)} = \frac{\beta}{2\gamma}\tan\alpha \approx 4.8\tan\alpha \qquad (z = 20.1\ \text{mm}). \tag{24}$$

At $\alpha = 10°$ this ratio is 0.84. From Section 5.3, with $\operatorname{tr} G^{-1} = 1/\beta^2 + 1/(4\gamma^2)$, a vertical axis costs $\operatorname{tr} G^{-1} - 1/(4\gamma^2) = 1/\beta^2$ and a horizontal axis costs $\operatorname{tr} G^{-1} - 1/(2\beta^2) = 1/(2\beta^2) + 1/(4\gamma^2)$ per b^2. Their ratio is $\frac{1}{2} + (\beta/2\gamma)^2 \approx 23$ at 20.1 mm.

6 From currents to drive settings

Amplitude. Each H-bridge switches the coil between $+V$ and $-V$ at the drive frequency, and a 20 kHz carrier gates it with duty D. The RLC filters out the carrier, so the coil sees a square

wave of height DV. A square wave of height A has fundamental amplitude $\frac{4}{\pi}A$, and the RLC also attenuates the higher harmonics, so

$$|I| = \frac{4DV}{\pi|Z(f)|}, \qquad |Z(f)| = \sqrt{R_s^2 + \Big(\omega L_s - \frac{1}{\omega C_s}\Big)^2}, \quad \omega = 2\pi f \qquad \Longrightarrow \qquad D_k = \frac{\pi|Z(f)|\,|I_k|}{4V g_k}, \tag{25}$$

where g_k is a per-channel gain trim, 1 until fitted. With $V = 12\,\mathrm{V}$, $R_s = 6.9\,\Omega$, $L_s = 1.4\,\mathrm{mH}$ and $C_s = 800\,\mu\mathrm{F}$, the nominal resonance is $f_0 \approx 150\,\mathrm{Hz}$; it has not been measured on the current capacitor bank. At f_0, $|Z| = R_s$ and $D = 100\%$ gives $|I| = 2.2\,\mathrm{A}$, so near resonance the $2\,\mathrm{A}$ continuous limit binds before the duty limit.

Saturation. If any channel exceeds 100% duty or $I_{cont} = 2\,\mathrm{A}$, the library scales the whole set by

$$s = \min\Big(1,\ \frac{100\%}{\max_k D_k},\ \frac{I_{cont}}{\max_k |I_k|}\Big). \tag{26}$$

$\mathbf{I} \to s\mathbf{I}$ gives $\mathbf{u} \to s\mathbf{u}$ and $\mathbf{v} \to s\mathbf{v}$, so $\hat{\mathbf{n}}$ and circularity are unchanged and the field becomes sb. The library reports s as `fieldFraction`. Section 7.3 gives the policy flight needs instead.

Phase. The firmware delays channel k by $\varphi_k/360^\circ$ of a period, so its phase setting is a lag, of opposite sign to $\arg(I_k)$. With a per-channel offset o_k,

$$\varphi_k = o_k - \arg(I_k) \pmod{360^\circ}. \tag{27}$$

The offsets absorb winding direction, wiring, and any phase the drive electronics add. The phase pattern predicted from the azimuths alone does not spin the rotor, and the empirical `SPIN_PHASES` does, so we fit o_k in the constructor: $R = \mathrm{Id}_3$ at $(0, 0, 15)\,\mathrm{mm}$ must send exactly `SPIN_PHASES`. The fit is $o = (90^\circ, 180^\circ, 180^\circ, 270^\circ)$. A reversed winding appears as a 180° offset. We cannot yet explain the 90° parts, so we treat the tilt direction as unverified until we run a step test on the rig.

Calibration errors. We suppose the real currents are $\kappa_k I_k$ for some complex κ_k, an error in impedance, gain or phase. If all four share one κ, the field is

$$\mathrm{Re}\big(B\,\kappa\mathbf{I}\,e^{\mathrm{j}\theta}\big) = |\kappa|\ \mathrm{Re}\big(B\mathbf{I}\,e^{\mathrm{j}(\theta+\arg(\kappa))}\big) = |\kappa|\,\mathbf{B}(\theta + \arg(\kappa)), \tag{28}$$

the commanded field scaled by $|\kappa|$ and shifted in time, with the same axis. If the κ_k differ, the field is $\mathrm{Re}(B\,\mathrm{diag}(\kappa)\mathbf{I}e^{\mathrm{j}\theta})$, which in general has a different axis. So the RLC values set the field magnitude only, and the axis depends only on the channel-to-channel trims g_k and o_k.

7 Algorithm

Algorithm 1 is `VectorField::settingsFor`. It never refuses with an error. Every input gets a defined output, and the outputs that make no field are all $D_k = 0$. Channels are indexed $k = 1, \ldots, 4$ (A, B, C, D), and every $\max_k$ is over those four.

Algorithm 1 `VectorField::settingsFor`$(\mathbf{r}, R, b, f)$

Require: point $\mathbf{r} \in \mathbb{R}^3$ (m, array frame); orientation $R \in \mathbb{R}^{3\times3}$ or quaternion $q \in \mathbb{R}^4$; field radius $b \in \mathbb{R}$ (T, signed); drive frequency f (Hz)

Require: per-channel gain trim g_k (1 until fitted) and phase offset o_k (deg, fitted to `SPIN_PHASES`, Section 6)

Ensure: carrier duty D_k (%), phase lag φ_k (deg), field fraction $s \in [0, 1]$

if a quaternion was given **then**
  if $\|q\| = 0$ **then**
    return $D_k = 0$, $s = 0$ ▷ no orientation: the caller keeps its previous drive
  end if
  $R \leftarrow R(q/\|q\|)$ ▷ every nonzero q gives a rotation
else if $RR^\top \neq \mathrm{Id}_3$ or $\det(R) \neq +1$ (each to 10^{-3}) **then**
  return $D_k = 0$, $s = 0$ ▷ a stretch or a mirror: the caller keeps its previous drive
end if
if b is not finite **then**
  return $D_k = 0$, $s = 0$ ▷ NaN or ∞ from the wire
end if
$\hat{\mathbf{e}}_1 \leftarrow R\hat{\mathbf{x}}, \quad \hat{\mathbf{e}}_2 \leftarrow R\hat{\mathbf{y}}$ ▷ Section 4
$B \leftarrow [\,\mathbf{b}_A\ \mathbf{b}_B\ \mathbf{b}_C\ \mathbf{b}_D\,]$ at $\mathbf{r}$ ▷ dipole sum, Section 2; T/A
$G \leftarrow BB^\top$ ▷ 3×3 Gram matrix
if $|\det(G)| \leq 10^{-40}$ (T/A)^6 **then**
  return $D_k = 0$, $s = 0$ ▷ B has rank < 3 at $\mathbf{r}$
end if
$G^{-1} \leftarrow \mathrm{adj}(G)/\det(G)$
$\mathbf{x} \leftarrow B^\top G^{-1}(b\,\hat{\mathbf{e}}_1), \quad \mathbf{y} \leftarrow B^\top G^{-1}(b\,\hat{\mathbf{e}}_2)$ ▷ minimum heat, Section 5.2
$I_k \leftarrow x_k - \mathrm{j}y_k, \quad |I_k| \leftarrow \sqrt{\mathrm{Re}(I_k)^2 + \mathrm{Im}(I_k)^2}$ ▷ A
if $f \leq 0$ **then**
  return $D_k = 0$, $s = 0$ ▷ C_s blocks DC: $|Z(f)| \to \infty$
end if
$\omega \leftarrow 2\pi f, \quad |Z(f)| \leftarrow \sqrt{R_s^2 + (\omega L_s - 1/(\omega C_s))^2}$ ▷ Ω
$D_k \leftarrow 100\% \cdot \pi\,|Z(f)|\,|I_k|\,/\,(4Vg_k)$ ▷ Section 6
if $\max_k |I_k| > 0$ **then**
  $s \leftarrow \min\big(1,\ 100\%/\max_k D_k,\ I_{cont}/\max_k |I_k|\big)$
else
  $s \leftarrow 1$ ▷ $b = 0$: every $D_k = 0$, so the coils are off
end if
$D_k \leftarrow s\,D_k$
$\arg(I_k) \leftarrow \mathrm{atan2}\big(\mathrm{Im}(I_k),\ \mathrm{Re}(I_k)\big)$ ▷ angle of I_k from the positive real axis
$\varphi_k \leftarrow \big(o_k - \arg(I_k)\big) \bmod 360°$, in $[0°, 360°)$
return D_k, φ_k for $k = 1, \ldots, 4$, and s

The safe outputs. Two kinds of input make no field.

- An orientation that cannot be used ($q = 0$, or $R \notin SO(3)$), a non-finite b, or a singular B returns $s = 0$. The firmware treats $s = 0$ as "do nothing" and keeps the previous drive.
- $f \leq 0$ also returns $s = 0$. This is the limit of the formula, because the series capacitor blocks DC, so $|Z| \to \infty$, $D_k \to \infty$ and $s \to 0$. `freq=` already rejects $f \leq 0$, so this case cannot happen in FLIGHT.

The field radius b is signed. The solve is linear in b, so $-b$ gives $\mathbf{I} \to -\mathbf{I}$, which leaves each $|I_k|$ the same and adds 180° to each $\arg(I_k)$. That is the field $\mathbf{B}(\theta + \pi)$, the same as a twist $R\,R_z(\pi)$. So a sign flip in flight is a 180° phase step under a locked rotor, which Section 4 says the rotor will not survive. $b = 0$ returns $D_k = 0$ with $s = 1$, and the firmware applies it, so the coils go off. The FLIGHT-only command

`vec=<x_mm>:<y_mm>:<z_mm>:<qw>:<qx>:<qy>:<qz>:<b_mT>`

calls this. The firmware stores the target and re-solves on every `freq=`, because the duties depend on $|Z(f)|$ and the axis does not.

8 Reachable workspace

8.1 Maximum field for a given axis

From $I_k = b\,\mathbf{p}_k \cdot (\hat{\mathbf{e}}_1 - \mathrm{j}\hat{\mathbf{e}}_2)$,

$$|I_k|^2 = b^2\big[(\mathbf{p}_k \cdot \hat{\mathbf{e}}_1)^2 + (\mathbf{p}_k \cdot \hat{\mathbf{e}}_2)^2\big] = b^2\,\mathbf{p}_k^{\top}(\mathrm{Id}_3 - \hat{\mathbf{n}}\hat{\mathbf{n}}^{\top})\,\mathbf{p}_k = b^2\,|\mathbf{p}_k \times \hat{\mathbf{n}}|^2. \tag{29}$$

The middle step uses $\hat{\mathbf{e}}_1\hat{\mathbf{e}}_1^{\top} + \hat{\mathbf{e}}_2\hat{\mathbf{e}}_2^{\top} = \mathrm{Id}_3 - \hat{\mathbf{n}}\hat{\mathbf{n}}^{\top}$; the last uses $|\mathbf{p}|^2 - (\mathbf{p}\cdot\hat{\mathbf{n}})^2 = |\mathbf{p}\times\hat{\mathbf{n}}|^2$. The twist does not appear. Requiring $|I_k| \le I_{lim}$ on every channel gives

$$\boxed{\,b_{max}(\mathbf{r}, \hat{\mathbf{n}}) = \frac{I_{lim}}{\max_k |\mathbf{p}_k \times \hat{\mathbf{n}}|}\,} \tag{30}$$

where I_{lim} is the smaller of I_{cont} and the current at 100% duty. Let δ_k be the angle between $\hat{\mathbf{n}}$ and $\mathbf{p}_k$, so $|\mathbf{p}_k \times \hat{\mathbf{n}}| = |\mathbf{p}_k| \sin\delta_k$. For a required b, channel k allows the axes with $\sin\delta_k \le I_{lim}/(b|\mathbf{p}_k|)$, a double cone about $\pm\mathbf{p}_k$. The reachable axes are the intersection of the four double cones.

On the array axis, a vertical axis gives $|\mathbf{p}_k \times \hat{\mathbf{z}}| = 1/(2\beta)$ for every k, so $b_{max} = 2\beta I_{lim}$. For the horizontal axis $\hat{\mathbf{n}} = \hat{\mathbf{x}}$, $|\mathbf{p}_k \times \hat{\mathbf{x}}|^2 = |\mathbf{p}_k|^2 - p_{k,x}^2$, which is largest for B and D:

$$b_{max}(\hat{\mathbf{x}}) = I_{lim} \Big/ \sqrt{\frac{1}{4\beta^2} + \frac{1}{16\gamma^2}}. \tag{31}$$

At 20.1 mm these give 6.8 and 1.4 mT. Table 1 evaluates b_{max} over height.

z (mm)	vertical b_{max} (mT)	horizontal b_{max} (mT)	largest tilt reachable at $b =$ 2 mT	3 mT	4.5 mT
10	6.5	5.7	90°	90°	90°
15	7.3	4.5	90°	90°	90°
18	7.2	2.8	90°	44.5°	15°
20	6.9	1.5	34°	16.5°	6.5°
25	5.9	1.2	24°	11°	3.5°
30	4.8	2.3	90°	21°	2°

Table 1: On the array axis, 150 Hz, dipole model, $I_{lim} = 2$ A. "Largest tilt" is the half-angle of the largest cone about $\hat{\mathbf{z}}$ whose every axis reaches b.

The horizontal b_{max} has a minimum between 20 and 25 mm, set by the $1/(16\gamma^2)$ term. Section 7.2 derives where γ vanishes.

8.2 The singular shell

On the array axis $\sigma_3 = 2\gamma$. A $+\hat{\mathbf{z}}$ dipole at lateral distance ρ makes, at height z on the array axis, with $|\mathbf{d}|^2 = \rho^2 + z^2$ and $\mathbf{m} \cdot \mathbf{d} = mz$,

$$B_z \propto \frac{3z^2}{|\mathbf{d}|^5} - \frac{1}{|\mathbf{d}|^3} = \frac{3z^2 - (\rho^2 + z^2)}{(\rho^2 + z^2)^{5/2}} = \frac{2z^2 - \rho^2}{(\rho^2 + z^2)^{5/2}}, \tag{32}$$

which is negative for $z < \rho/\sqrt{2}$ and positive above. Each channel's inner coils sit at $\rho = \sqrt{26.5^2 + 10.5^2} = 28.5$ mm and change sign at 20.2 mm. Its outer coils sit at $\rho = \sqrt{47.5^2 + 10.5^2} = 48.6$ mm and change sign at 34.4 mm. Between those heights the two pairs have opposite sign, and their sum γ is zero at 22.4 mm. Off the array axis we find the zero of σ_3 numerically:

x (mm)	0	6	10	15	20
z where $\sigma_3 = 0$, dipole model (mm)	22.4	22.0	21.2	19.4	16.7
z where $\sigma_3 = 0$, exact loop model (mm)	20.8	20.3	19.4	17.5	15.0

Where $\sigma_3 = 0$, B has rank 2 and every field it makes lies in the plane of $\mathbf{a}_1$ and $\mathbf{a}_2$. A circular field must then have $\mathbf{u}$ and $\mathbf{v}$ in that plane, so $\hat{\mathbf{n}} = \pm\mathbf{a}_3$, which is $\pm\hat{\mathbf{z}}$ on the array axis. This holds for any current. Near the shell the solve succeeds, but $\mathbf{p}_k$ contains $1/(4\gamma)$, so the common mode grows without bound and s shrinks the field toward zero.

The two models place the shell 1.6 mm apart, and the real coils are 22 mm-long solenoids, which neither model describes. We have to measure the height on the rig, by finding where a commanded tilt stops tilting.

8.3 Consequences for planning

The rotor stays phase-locked only if $b \geq b_{lock}(f) = 1.0\,\text{mT}\,(f/110\,\text{Hz})^2/0.8$ (theory.md §12.3: drag torque grows as f^2, with a 0.8 safety factor on the lock angle). That is 1.05 mT at 101 Hz and 4.56 mT at 210 Hz. A pose $(\mathbf{r}, \hat{\mathbf{n}})$ is feasible at frequency f when $b_{lock}(f) \leq b_{max}(\mathbf{r}, \hat{\mathbf{n}})$. From this:

1. We cross the shell with the axis vertical. A climb through it must bring the tilt in before the band and restore it after, so the planner must look ahead along the trajectory.
2. In flight, we reduce tilt before amplitude. Scaling b below b_{lock} makes the rotor slip. We keep $b \geq b_{lock}$ and move $\hat{\mathbf{n}}$ along the great circle toward $\hat{\mathbf{z}}$ until $b_{max} \geq b$; b_{max} is closed-form, so this is a one-dimensional bisection. The library's uniform scaling does the opposite.
3. We limit the axis slew rate. The rotor's axis follows commands at about 0.5 Hz (theory.md §12.6).

Underactuation. Away from the shell there are eight real unknowns and six field components, so the field at a point is fully actuated. With the field fixed, the remaining two unknowns are the complex amplitude of the null mode, which cannot set all three components of the gradient force. Lateral position must therefore be controlled through thrust tilt, and the Earnshaw anti-trap (theory.md §12.6) is a disturbance set by the field choice.

9 Verification and open items

- `lib/VectorField/check/self_check.cpp` checks: the geometry against the GUI (4.58 and 4.29 mT); $\sigma = (2.42, 2.42, 0.36)$ mT/A at 20.1 mm and the null mode; 200 random orientations

reached to 10^{-3} deg with a circular field; $R = \mathrm{Id}_3$ sending `SPIN_PHASES`; a twist changing only the phases; saturation keeping the axis; safe outputs for bad input (signed b, $b = 0$, $f = 0$, $q = 0$, non-rotations).

- The C++ matched the original Python solver to float precision on 40 cases.
- Open, all needing the rig: the tilt direction (the offsets o_k), the shell height, f_0, and the gains g_k.